SQL Quick Reference

Part 1: Building Tables and Changing Data

Each command below has a short explanation and an example. The examples build and use a simple students table.

1. CREATE TABLE

Creates a new table. Inside the parentheses, list each column with its data type, then the keys.

Example:

```
CREATE TABLE students
(   id       INT         NOT NULL,
    name     VARCHAR(30) NOT NULL,
    age      INT,
    major    VARCHAR(20),
    credits  INT,
    email    VARCHAR(40),
    PRIMARY KEY (id) );
```

2. NOT NULL

Means a column must always have a value. Leaving it empty causes an error.

Example (one column line inside CREATE TABLE):

```
name  VARCHAR(30)  NOT NULL,
```

3. PRIMARY KEY

Names the column that uniquely identifies each row. No two rows can share the same value, and it cannot be empty.

Example (the last line inside CREATE TABLE):

```
PRIMARY KEY (id)
```

4. FOREIGN KEY ... REFERENCES

Links a column to the primary key of another table. Every value must match a row that already exists in that table.

Example:

```
CREATE TABLE enrollments
(   enrollid   INT         NOT NULL,
    studentid  INT         NOT NULL,
    course     VARCHAR(10) NOT NULL,
    PRIMARY KEY (enrollid),
    FOREIGN KEY (studentid) REFERENCES students(id) );
```

5. ALTER TABLE

Changes a table that already exists, for example by adding or removing a column.

Example:

```
ALTER TABLE students ADD COLUMN phone VARCHAR(12);
ALTER TABLE students DROP COLUMN phone;
```

6. DROP TABLE

Deletes a whole table, including all of its rows. Drop a table that points to another table first, so enrollments is dropped before students.

Example:

```
DROP TABLE enrollments;
```

7. INSERT INTO ... VALUES

Adds a new row to a table. The values go in the same order as the columns, with text inside single quotes.

Example:

```
INSERT INTO students
VALUES (1, 'Maya', 18, 'Biology', 15, 'maya@example.edu');
```

8. UPDATE ... SET

Changes values in rows that already exist. WHERE picks the rows to change; without it, every row changes.

Example:

```
UPDATE students
SET major = 'Chemistry'
WHERE id = 1;
```

9. DELETE FROM

Removes rows from a table. WHERE picks the rows to remove; without it, every row is removed.

Example:

```
DELETE FROM students
WHERE id = 4;
```